OpenJFEM: Origins of the Numerical Methods

A Literature Review Mapping the Reference Corpus to the Solver Pipeline

JFEM

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Abstract

This report traces the numerical methods implemented in OpenJFEM ("JFEM") back to their sources in the open literature. It is built from two inputs: the curated corpus of reference papers under `02_DEVELOPMENT_AND_VALIDATION/External_papers` (forty-two peer-reviewed papers and one monograph, plus four Reference manuals), and a direct audit of the solver source tree under `src/`. The review is organised along the JFEM *analysis pipeline*—parsing, model building, element formulation, global assembly, solution sequences, and result recovery—so that each implemented technique is presented next to the paper(s) that originated it and the file in which JFEM realises it. The recurring design philosophy is parity with the Reference CQUAD4/CTRIA3 shell family [1, 24], reached through the mixed-interpolation and residual-bending-flexibility ideas of Dvorkin and Bathe [3, 4] and MacNeal [1], with discrete-shear (DKMQ), Hu–Washizu, drilling-rotation, and geometrically-exact formulations held in reserve as research kernels.

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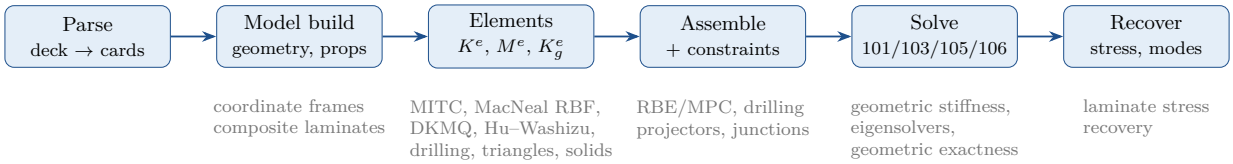


Figure 1: The JFEM analysis pipeline and where each corpus theme enters. The element stage is the densest, reflecting that shell-element technology is the heart of the solver.

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1 Scope and method

JFEM is a Julia finite-element solver for bulk-data decks that supports linear statics (SOL 101), normal modes (SOL 103), linear buckling (SOL 105), geometric nonlinear statics (SOL 106), and a sizing-optimisation subset (SOL 200-lite). Its defining engineering goal is *quantitative parity* with Reference on shell-dominated airframe models. That goal dictates which methods were adopted: where a choice existed, JFEM follows the formulation that Reference's CQUAD4/CTRIA3 elements use, and reserves more elaborate academic elements (Hu–Washizu mixed shells, DKMQ, geometrically-exact stress-resultant shells) as gated research kernels for comparison.

This document connects *papers* to *code*. For every method family we state: the originating reference(s); the numerical idea; and the JFEM file and function that implements it. Implementation locations were verified by reading the source; the production element kernels live in `src/FEMKernels.jl`, the solution-sequence logic in `src/JFEMSolver.jl` and `src/solver/`, model construction in `src/ModelBuilder.jl`, and the deck parser in `src/parsing/`. Experimental kernels are quarantined under `src/experimental/` and reached only through environment switches; they are labelled *research* below.

The corpus. The forty-six reference documents are grouped into thirteen thematic folders that map almost one-to-one onto the element and solution technology in JFEM. Figure 1 shows the pipeline and the stage at which each corpus theme enters.

2 Parsing and model building

2.1 Bulk-data parsing and coordinate frames

The parser (`src/parsing/`) reads fixed-, large-, free-field and tab-delimited cards, resolves nested `INCLUDE` statements (`card_processor.jl`), and builds the case-control/bulk-data split. Rectangular, cylindrical, and spherical coordinate systems (`CORD2R`, `CORD2C`, `CORD2S`) and element material frames (`MCID`, `THETA`) are resolved in `src/parsing/extract_geometry.jl` and applied during element-local frame construction in `FEMKernels.jl`. These are Reference data-model conventions; the authoritative references are the Reference manuals [44, 45, 47].

2.2 Composite laminates: the constitutive layer

JFEM supports isotropic shells through `PSHELL`/`MAT1` and laminated composites through `PCOMP`/`MAT8`. The model builder assembles the classical laminate A, B, D (and transverse-shear) stiffness resultants by through-thickness integration of the transformed ply stiffnesses $\bar{Q}_k$—the standard *classical laminated plate theory* (CLPT) and *first-order shear deformation theory* (FSDT) construction. The canonical source for this constitutive layer, including ply integration, the ABD matrices, and shear-correction factors, is Reddy's monograph [31]. JFEM uses the equivalent-single-layer,

resultant (D-matrix) form on its production shells; ply-by-ply through-thickness integration exists only in the research MITC4-3D kernel.

Three further corpus papers inform the composite capability rather than the default code path. Khoa and Thinh [33] give a higher-order shear deformation (HSDT) element that removes the need for a shear-correction factor through a parabolic through-thickness shear field; Baskin's 3-D elasticity study [32] supplies independent solid-model stress benchmarks; and Rolfes, Rohwer and Ballerstaedt [34] provide an efficient, equilibrium-based post-processing scheme for recovering transverse normal and interlaminar stresses from an FSDT solution—a natural extension point for JFEM's laminate stress recovery (§6).

3 Element formulation: the shell core

Shell-element technology is where JFEM concentrates its modelling effort and where the corpus is richest. The production quadrilateral and triangle target Reference CQUAD4/CTRIA3 behaviour; several alternative formulations are implemented as research kernels for comparison and future calibration.

3.1 The mixed-interpolation (MITC) lineage

The transverse-shear treatment in JFEM's quadrilateral descends directly from the *mixed interpolation of tensorial components* (MITC). Dvorkin and Bathe [3] introduced the original four-node continuum-mechanics shell with assumed covariant transverse-shear strains tied at the four edge midpoints, curing shear locking with no spurious zero-energy modes; Bathe and Dvorkin [4] gave the flat-plate specialisation (the MITC4 plate). This assumed-shear field—sampling shear at the edge midpoints and interpolating the covariant components in the natural coordinate system—is the locking cure used in JFEM's default quad kernel (`FEMKernels.jl`, `stiffness_quad4_default_generic`, covariant shear B-matrix `Bs_cov`).

The MITC family was evaluated systematically by Bathe, Iosilevich and Chapelle [5], who established the strain-energy convergence framework, the membrane- vs. bending-dominated asymptotic categories, and the inf–sup condition that a thickness-robust shell must satisfy—the theoretical yardstick JFEM's validation suite uses. The modern "plus" elements refine the *membrane* field: Ko, Lee and Bathe [6, 7] trace membrane locking to out-of-plane element distortion and construct an assumed membrane strain (from a distortion vector and characteristic geometry vectors) that vanishes in pure bending yet alleviates locking on curved or distorted meshes; the companion 2-D-solid paper [8] reuses that membrane field, and the nonlinear paper [9] carries it into large-rotation analysis with quaternion director updates. The benchmark compendium of Ko *et al.* [10] and the triangular modal study of Lee *et al.* [11] provide the obstacle-course problems (pinched cylinder, Scordelis–Lo roof, hemisphere, hyperboloid) JFEM validates against. JFEM ships an MITC4-style production path and a research *MITC4-3D* director-based degenerate-solid kernel (`src/experimental/mitc4_3d_kernel.jl`), the latter explicitly motivated in its comments by the Ko–Lee–Bathe curved-shell work.

3.2 The MacNeal CQUAD4 and the production default

Although the MITC assumed-shear field defines the kinematic skeleton, the *default* production quad in JFEM matches Reference's CQUAD4 through MacNeal's *residual bending flexibility* (RBF) correction [1]. MacNeal's insight—first shown on a two-node beam—is that the bending energy lost by a low-order element (the missing cubic curvature term) can be recovered by *augmenting the transverse-shear flexibility* with a residual term $\ell^2/12EI$, while sampling shear strain at the element centre by reduced integration. JFEM implements exactly this in `FEMKernels.jl` (`add_quad4_macneal_shear_rbf!`, the production default selected by `JFEM_Q4_KERNEL=macneal`); the code comments cite MacNeal (1978) and record a calibration factor that reproduces Reference's

per-element stiffness to within a few percent across MAT1/MAT8/PCOMP probes. The Reference membrane half of the element—corner *drilling* rotations built by condensing an eight-node element, with spurious-mode penalty parameters—comes from MacNeal and Harder [24] (§3.5). The companion NASA implementation note of Potvin and Leick [2] documents a distinct degenerated curved CQUAD4 and is retained for historical context; JFEM follows the MacNeal RBF line, not the degenerated-continuum one, for its default.

3.3 Discrete-shear quadrilaterals (DKMQ) — research

A parallel, mathematically elegant route to a locking-free Reissner–Mindlin quad is the *discrete-shear* family. Its conceptual root is Tessler and Hughes [12], who interpolate the deflection one order higher than the rotations (interdependent variable interpolation) and introduce an element-appropriate shear-correction factor, excluding shear locking without selective integration. Katili *et al.* [13] cast DKQ, DSQ, MITC4, and DKMQ in a single modified Hu–Washizu framework, making explicit how each assumes the shear field; the DKMQ element's ϕ_k factor smoothly morphs the element between Reissner–Mindlin and Kirchhoff–Love limits, giving distortion tolerance. The DKMQ shell was extended to warped geometry and composites as DKMQ24 [14], to improved membrane behaviour with Allman drilling rotations [15], and to a 2025 discrete-shear-projection plate [16]. JFEM contains flat-plate DKQ/DKMQ kernels in `src/experimental/dkq_dkmq_flat_plate_kernels.jl` (gated, research), with comments citing the Katili–Maknun–Batoz–Ibrahimbegović 2018 composite DKMQ24 paper.

3.4 Hu–Washizu and assumed-stress hybrids — research

For coarse and distorted meshes, multi-field variational elements offer the strongest locking cures. JFEM carries a research suite in `src/experimental/hu_washizu_kernel.jl` spanning Pian-type hybrid-stress membrane, assumed-moment bending, and full three-field Hu–Washizu kernels. The relevant lineage in the corpus has two strands. The *assumed-stress* strand: Yuan and Pian [17] establish reference (geodesic skew) coordinates and distortion measures that keep a hybrid membrane element's assumed stresses linear in physical coordinates under distortion, and Li, Xie and Zhang [18] prove the Pian–Sumihara element is λ-uniformly (incompressibility-) convergent. The *three-field Hu–Washizu* strand: Wagner and Gruttmann [19] build a mixed shell with quadratic strain enrichment orthogonal to the constant part; Gruttmann and Wagner [20] reduce it to a one-point-quadrature element with analytic, parameter-free hourglass stabilisation; Daszkiewicz *et al.* [21] extend the principle to nonlinear Cosserat (drilling) shells with asymmetric strain/stress fields; and Wisniewski and Turska [22] catalogue reduced assumed-field representations for solid-shell elements, combining ANS, EAS and a MacNeal-style RBF shear correction. These are not on the default path—JFEM's comments record that a 2026 Hu–Washizu trial regressed against the calibrated MacNeal kernel—but they document the design space and are the basis of the high-performance shell route in the TACS-formulation backend.

3.5 Drilling rotations and projectors

A true six-degree-of-freedom shell node requires a stiffness for the in-plane (*drilling*) rotation about the shell normal. JFEM adds a drilling contribution in every shell kernel (the `Bd` drilling B-matrix and the `k6rot` penalty in `FEMKernels.jl`), the same device Bathe and Ho [28] used (a small fictitious $K_{\theta z}$) to remove the in-plane-rotation singularity in a flat shell. The rigorous variational foundation is Hughes and Brezzi [23], who show that naive equal-order interpolation of an independent rotation field fails the discrete ellipticity condition and supply the regularising γ^{-1} penalty that makes drilling elements stable and convergent. The practical membrane construction—Allman/Cook corner rotations with spurious-mode penalty parameters—is MacNeal and Harder [24], which is precisely the membrane half of the Reference QUAD4 that JFEM targets.

The corpus also includes the *projector* technology of Rankin and Nour-Omid [25] and Nour-Omid and Rankin [26]: an element-independent corotational projector that filters rigid-body modes, cures the warping sensitivity of flat elements on curved surfaces, and provides a consistent tangent for large rotations. Belytschko *et al.* [27] give the unifying stress-projection view of membrane and shear locking (and the DKT+CST flat triangle). JFEM does not yet use an explicit corotational projector on its production path; these references define the route for robust large-rotation and warped-element behaviour and inform the SOL 106 and backend work.

3.6 Triangular shells (CTRIA3)

JFEM implements a three-node shell (`stiffness_tria3` in `FEMKernels.jl`). The classical template is the flat facet triangle of Bathe and Ho [28]: a constant-strain-triangle membrane plus a *discrete Kirchhoff* (DKT) bending stiffness plus a fictitious drilling stiffness, with an updated-Lagrangian nonlinear option. The DKT bending engine itself is the subject of the definitive comparative study by Batoz, Bathe and Ho [29], which derives the DKT element's compact, exactly integrated stiffness and finds DKT and the hybrid-stress model the most reliable nine-DOF triangular plate elements. For the modern mixed-interpolation triangle, Lee *et al.* [11] analyse the MITC3+ element's modal behaviour and give its assumed-shear field and tying distance—the reference for a locking-free Reissner–Mindlin triangle should JFEM move from the DKT-style facet to a full MITC3+ formulation.

3.7 Solid elements and distortion

JFEM provides solid element stiffness and mass for tetrahedra, hexahedra and pentahedra (`stiffness_tetra4`, `stiffness_hexa8`, the `consistent_mass_*` and `reference_lumped_mass_*` routines in `FEMKernels.jl`). The corpus contributes one specialised reference on robustness: Zhou *et al.* [30] construct an unsymmetric eight-node hexahedron whose trial functions are analytical (Trefftz) solutions in oblique coordinates, giving high tolerance to severe mesh distortion and circumventing MacNeal's theorem. This is a design target for distortion-insensitive bricks rather than the current isoparametric implementation.

4 Global assembly and constraints

After element matrices are formed, JFEM assembles the global system and applies multipoint and rigid constraints (`src/solver/constraints.jl`): rigid elements `RBE2`, `RBE3`, `RBE1`, splines `RSPLINE`, explicit `MPC` equations, and automatic singularity suppression (`AUTOSPC`). These follow the Reference definitions and numerical conventions [44, 45]; the rigid-element and constraint elimination behaviour is calibrated for parity rather than drawn from a single research paper.

Two corpus themes bear on assembly correctness for compound structures. The drilling rotation (§3.5) is what permits well-posed assembly of *coplanar* flat shells and shell-to-beam connections without a rotational singularity. More broadly, Pietraszkiewicz and Konopińska [43] review the kinematic and dynamic *jump conditions* that must hold at shell junctions, branches, and stiffener intersections, and explain why six-parameter (drilling-DOF) shell models are advantageous there—directly relevant to the stiffened-panel models that JFEM's web application targets.

5 Solution sequences

5.1 Linear statics (SOL 101)

The static path solves $Ku = f$ after constraint elimination, with stress and reaction recovery. The element stiffness underlying it is the MITC/MacNeal shell of §3.1–§3.2; the solver conventions follow the Reference references [44, 47].

5.2 Normal modes (SOL 103)

The modal path forms a generalised eigenproblem $K\phi = \lambda M\phi$. JFEM assembles both lumped (bulk-data diagonal) and consistent mass matrices (`reference_lumped_mass_*` and `consistent_mass_*` in `FEMKernels.jl`) and uses a dense symmetric-definite eigensolver for the free system. The numerical-methods conventions (mass options, eigenvalue extraction) track the Reference numerical methods guide [46]. The modal behaviour of a mixed-interpolation element is analysed by Lee *et al.* [11], whose mode-level diagnostics (zero transverse shear in bending modes) are the kind of check JFEM uses to confirm absence of spurious modes.

5.3 Linear buckling (SOL 105)

Buckling is JFEM's historical production focus. It solves the generalised eigenproblem $(K + \lambda K_g)\,\phi = 0$, where the geometric (stress) stiffness K_g is assembled from the static pre-stress state (`src/solver/assembly.jl`, `src/JFEMSolver.jl`, with the SOL 105 logic in `src/solver/solve_case.jl`). The geometric-stiffness construction for a discrete-shear quad is given explicitly by Wong *et al.* [36], who augment the DKMQ Hu–Washizu functional with the membrane pre-stress strain energy; Katili *et al.* [37] extend the eigenvalue buckling formulation (with von Kármán membrane strains) to functionally-graded sandwich plates and supply benchmark critical loads against ABAQUS S4R and analytical solutions. The DKMQ buckling element of [35] is the direct reference for the geometric-stiffness recipe and the buckling benchmark factors JFEM validates against; the eigenvalue extraction conventions follow [46].

5.4 Geometric nonlinear statics (SOL 106)

JFEM has a SOL 106 path (`src/solver/solve_case.jl`, `src/JFEMSolver.jl`) for geometrically nonlinear response. The theoretical backbone is the geometrically-exact, stress-resultant shell of Simo and Fox [38] (Part I: one-director Cosserat-surface formulation with a singularity-free director parametrisation on $S^2/SO(3)$ via the exponential map), with the linear computational treatment in Simo, Fox and Rifai [39] and the nonlinear implementation—exact director update and consistent tangent for Newton iteration and bifurcation analysis—in Part III [40]. The drilling-rotation geometrically nonlinear shell of Ibrahimbegović [41] and Ibrahimbegović and Frey [42] adds a consistent drilling rotation (rotation-vector parametrisation, incompatible-modes enhancement) so that folded and stiffened shells assemble correctly under large rotation. The corotational projector papers [25, 26] and the nonlinear MITC4+ [9] provide alternative element-independent and assumed-strain routes to the same end. These references define the formulation space for JFEM's nonlinear and TACS-formulation backend development.

6 Result recovery

Stress and force recovery (`src/solver/stress_recovery.jl`) returns element stresses, shell stress resultants, and SPC reactions, and exports a binary `.jfem` result file, JSON, a markdown report, and optional VTK/HDF5. For laminated composites, the through-thickness and interlaminar stress state is the quantity most sensitive to the recovery scheme; Reddy [31] defines the ply-stress recovery from laminate resultants, Rolfes *et al.* [34] give an efficient equilibrium-based scheme for transverse shear and normal stresses recoverable from an FSDT solution, and the HSDT and 3-D-elasticity studies [33, 32] supply reference through-thickness stress distributions for validation.

7 Synthesis: the JFEM design position

Read as a whole, the corpus explains a deliberate, parity-driven design.

- **Production shells** take the locking cure from the MITC assumed-shear field [3, 4] and the quantitative stiffness from MacNeal's residual bending flexibility [1], with the drilling membrane of MacNeal and Harder [24]—i.e. JFEM's default element is, by construction, an open re-derivation of the Reference CQUAD4/CTRIA3 family.
- **Triangles** follow the DKT facet template [28, 29].
- **Buckling**, the historical focus, uses the geometric-stiffness eigenproblem with discrete-shear references for the recipe [35, 36].
- **Research kernels**—DKMQ [13, 14], Hu–Washizu / hybrid stress [19, 20, 17], and geometrically-exact / drilling-rotation shells [38, 41]—are held in reserve to push accuracy on coarse and distorted meshes and to underpin the nonlinear and TACS-formulation backend.

Table 1 summarises the mapping from pipeline stage to method to primary reference.

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Table 1: Pipeline stage → method → primary source(s) and JFEM status.

Stage	**Method**	**Primary source(s)**	**JFEM**
Model	Composite laminate ABD (CLPT/FSDT)	Reddy [31]	Production
Element	MITC4 assumed shear	Dvorkin–Bathe [3]; Bathe–Dvorkin [4]	Production
Element	CQUAD4 residual bending flexibility	MacNeal [1]	Production (default)
Element	Drilling membrane (Allman/Cook)	MacNeal–Harder [24]; Hughes–Brezzi [23]	Production
Element	MITC4+ membrane (distortion)	Ko–Lee–Bathe [6, 7]	Reference
Element	DKMQ / DSQ discrete shear	Katili *et al.* [13, 14]	Research
Element	Hu–Washizu / hybrid stress	Wagner–Gruttmann [19]; Yuan–Pian [17]	Research
Element	CTRIA3 (DKT facet)	Bathe–Ho [28]; Batoz–Bathe–Ho [29]	Production
Element	MITC3+ triangle	Lee *et al.* [11]	Reference
Element	Distortion-tolerant hexahedron	Zhou *et al.* [30]	Reference
Assembly	Corotational projectors	Rankin–Nour-Omid [25, 26]	Reference
Assembly	Shell junctions / jump conditions	Pietraszkiewicz–Konopińska [43]	Reference
Solve 105	Geometric stiffness buckling	Wong–Katili [36]; [35, 37]	Production
Solve 106	Geometrically-exact shell	Simo–Fox(–Rifai) [38, 39, 40]	Developing
Solve 106	Drilling-rotation nonlinear shell	Ibrahimbegović(–Frey) [41, 42]	Reference
Recover	Laminate / interlaminar stress	Reddy [31]; Rolfes *et al.* [34]	Production/Ref.
All	Reference parity conventions	Reference manuals [44, 45, 46, 47]	Production